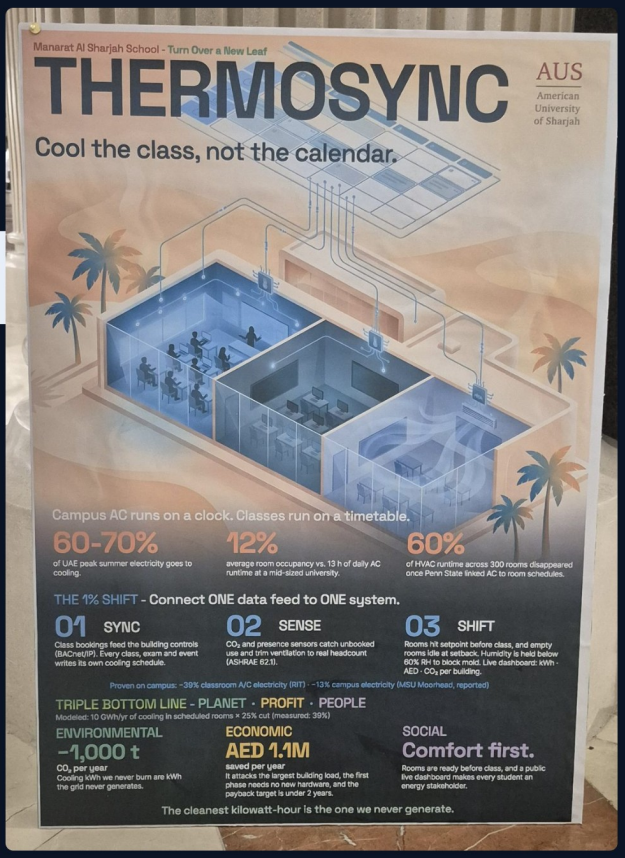
Sharakah × Net Impact Sustainability Challenge 2026, American University of Sharjah

THERMOSYNC

Cool the class, not the calendar.

Campus air conditioning runs on a fixed clock, so it cools empty rooms. THERMOSYNC links the class timetable to the building controls and uses sensors to confirm who is really there, so only rooms in use get cooled.

Project lead: [YOUR NAME]. Concept, systems design, one-minute pitch and A1 poster.



The presented A1 poster at the exhibition.

Campus AC runs on a clock. Classes run on a timetable.

60–70%

of UAE peak summer electricity goes to cooling.

U.S. Commercial Service, UAE market intelligence

12% vs 13 h

average room occupancy against daily AC runtime at a mid-sized university.

MSU Moorhead case study, vendor-reported

60%

of HVAC runtime across 300 rooms disappeared once Penn State linked AC to room schedules.

Penn State Office of the Physical Plant

The 1% shift: connect one data feed to one system.

01

02

03

Sync

Timetable to thermostat

Class bookings feed the building management system over BACnet/IP, the standard language buildings use. Every class, exam and event writes its own cooling schedule.

Sense

Verify, don't assume

CO_2 and presence sensors (motion plus mmWave radar, no cameras) catch unbooked use and match ventilation to real headcount, following ASHRAE 62.1.

Shift

Pre-cool, set back, repeat

Rooms hit setpoint before class, and empty rooms idle at setback. Humidity stays at or below 60% RH to block mold. A live dashboard shows kWh, AED and CO_2 per building.

Control priority: safety and manual override, then sensor presence, then the timetable, then default setback. If the service goes offline, the building reverts to its existing fixed schedule.

Rollout in three phases

Phase 1, Sync. No new field hardware, only a small server and building-system programming. It assumes the building system accepts remote schedule writes, which is still to be confirmed.

Phase 2, Sense. Sensors go into one pilot building and are measured against comparable control rooms.

Phase 3, Scale. Campus-wide rollout with the public dashboard.

Impact, modeled and not measured

Basis: 10 GWh a year of cooling in scheduled rooms, cut by a conservative 25%, saves 2.5 GWh. The 10 GWh is a placeholder for a campus's real figure, and 25% sits below the 39% measured at Rochester Institute of Technology.

−1,000 t

CO_2 per year at 0.4 kg per kWh (DEWA 2020 emission factor). Cooling kWh we never burn are kWh the grid never generates.

AED 1.1M

saved per year at AED 0.44 per kWh (SEWA commercial rate, secondary sources). Payback under two years is a design target, not a measured result.

Comfort first

Rooms are ready before class, and a public live dashboard lets every student follow the campus's energy use.

Evidence from other campuses: −39% classroom A/C electricity in a Rochester Institute of Technology experiment (classrooms only), and −13% campus electricity reported by MSU Moorhead (vendor-reported).

What this is and isn’t

- A concept with a modeled impact case. It has not been built or piloted.
- The evidence comes from US campuses and has not yet been shown in UAE humidity.
- The 39% result covers classrooms only, so campus-wide savings would be lower.
- There is no local energy data and no vendor cost quote yet, which is why the proposal starts with a pilot.
- The approach is proven elsewhere. The contribution is adapting it to hot, humid campuses with a humidity guardrail, sensor verification, a public dashboard and a no-new-hardware first step.

Sources

- U.S. Commercial Service, UAE design and construction market intelligence: cooling share of peak summer electricity.
- Minnesota State University Moorhead case study (EMS Software, vendor-reported): room occupancy, runtime and campus electricity.
- Penn State Office of the Physical Plant: runtime reduction across 300 rooms.
- Rochester Institute of Technology dissertation (Marozas-Aliaga, 2018): classroom A/C electricity.
- DEWA 2020 emission factor (0.4 kg CO_2 per kWh) and SEWA commercial rate (AED 0.44 per kWh).

How it was made

The concept, research and copy were developed with AI assistance, and the poster illustration was generated with AI. The team presented the one-minute pitch and answered the judges’ questions. Judges’ feedback highlighted the no-new-hardware first phase.